



Final Examinations

2nd Semester, S.Y. 2024 – 2025
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HAU School of Computing
First Year | Bachelor of Science in
Computer Science

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**Should there be any mistakes, inconsistencies, or inaccuracies in this material, please feel free to reach out to the transcriber. Good luck and study well for your exams!*

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Functions

Introduction to Functions

Functions are used to analyze the time needed to execute an algorithm. Their application in various fields ranges from representation of the computational complexity of algorithms, counting objects, study of sequences and strings, to name a few.

A **function** establishes a relationship between two sets, **A** and **B**, by assigning each element in set **A** to exactly one element in set **B**.

A function is denoted as **f: A → B**, which is read as "a function **f** from **A** to **B**."

- A is called the **domain** of **f**, while B is called the **co-domain** of **f**.
- If **f(x) = y**:
 - 'x' is called the **pre-image** of y under f
 - 'y' is called the **image** of x under f.

Capital letters (**A**, **B**) represent **sets**; small letters (*x*, *y*) represent **elements** in sets.

Key Properties of Functions

- Each element in **A** is mapped to **only one** element in **B**.
 - If an element in A is mapped to two or more elements in B, then it is a **non-function**.
 - Moreover, if an element in set A is NOT mapped to any element in set B, then it is also a **non-function**

- However, an element in **B** may correspond to multiple elements in **A** (i.e., have multiple pre-images).
- Functions can be **one-to-one** (injective) or **many-to-one**, but they cannot be one-to-many.

Range

- The **range** of a function **f**, denoted as **f(A)**, is the set of all images of elements in **A**.
- The **range** is always a subset of **B**, meaning that for every **y** in the range, there exists an **x** in **A** such that **y = f(x)**.

Example 1. Mapping Holidays to Their Dates

Let:

- A = {Christmas, New Year, Valentine's Day, Independence Day}
- B = {December 25, January 1, February 14, June 12}

Each holiday in **A** is assigned to exactly one date in **B**, forming a valid function.

Domain (A): {Christmas, New Year, Valentine's Day, Independence Day}

Co-Domain (B): {December 25, January 1, February 14, June 12}

Pre-Images and Images:

- Christmas → December 25
- New Year → January 1
- Valentine's Day → February 14
- Independence Day → June 12

Range of f

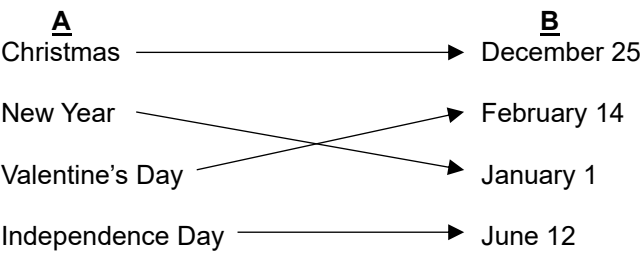
The **range** of a function consists of all output values (images) that result from applying **f** to elements in **A**.

In this case, the **range** of **f**, denoted as **f(A)**, is:

f(A) = {December 25, January 1, February 14, June 12}

Since every date in **B** is assigned to at least one holiday in **A**, the **range** of **f** is equal to the entire co-domain **B**.

The same relation can also be illustrated this way:



Example 2. Sets of Numbers

Let:

- A = {1, 2, 3, 4}
- B = {3, 4, 5, 6}
- C = {-2, -1, 0, 1, 2}
- D = {7, 3, 1, 5}
- E = {3, 7, 9}
- F = {0, 2, 4, 1}

Determine whether each of the following relations is a function:

a) f: A → B = {(1,3), (2,4), (3,5), (4,6)}

b) f: C → D = {(-2, 7), (-1,3), (0,1), (1,5), (2,5)}



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c) $f: E \rightarrow F = \{(3,0), (3,2), (7,4), (9,1)\}$

Answers:

- a) **Yes**, this is a function because each element in A has exactly one unique image in B.
- b) **Yes**, this is a function because each element in C is assigned to only one element in D.
- c) **No**, this is not a function because 3 in E is mapped to both 0 and 2, violating the definition of a function (one input $\rightarrow$ one output).

Identify the following:

- a) What is the domain of $f: A \rightarrow B$?
- b) What is the codomain of $f: C \rightarrow D$?
- c) What is the range of the function $f: A \rightarrow B$?
- d) What is the range of the function $f: C \rightarrow D$?
- e) What is the pre-image/s of 5 in $f: A \rightarrow B$?
- f) What is the pre-image/s of 5 in $f: C \rightarrow D$?
- g) What is the image of 2 or $f(2)$ in $f: A \rightarrow B$?

Answers:

- a) {1, 2, 3, 4}
- b) {7, 3, 1, 5}
- c) {3, 4, 5, 6}
- d) {7, 3, 1, 5}
- e) {3}
- f) {1, 2}
- g) 4

Cardinality

The cardinality of a set refers to the number of elements it contains. It is denoted as $|A|$, where A is a set.

In **Example 2** above, the following are the cardinalities of the given sets:

- o $|A| = 4$ ($A = \{1, 2, 3, 4\}$)
- o $|B| = 4$ ($B = \{3, 4, 5, 6\}$)
- o $|C| = 5$ ($C = \{-2, -1, 0, 1, 2\}$)
- o $|D| = 4$ ($D = \{7, 3, 1, 5\}$)
- o $|E| = 3$ ($E = \{3, 7, 9\}$)
- o $|F| = 4$ ($F = \{0, 2, 4, 1\}$)

Cartesian Product

The **Cartesian product** of two sets, $A \times B$, is the set of all ordered pairs (a,b), where $a \in A$ and $b \in B$.

The number of elements in $A \times B$ is given by $|A| \times |B|$.

In **Example 2** above, the following Cartesian Products may be derived:

$R(A \times B) = \{(1,3), (1,4), (1,5), (1,6), (2,3), (2,4), (2,5), (2,6), (3,3), (3,4), (3,5), (3,6), (4,3), (4,4), (4,5), (4,6)\}$

$R(C \times D) = \{(-2,7), (-2,3), (-2,1), (-2,5), (-1,7), (-1,3), (-1,1), (-1,5), (0,7), (0,3), (0,1), (0,5), (1,7), (1,3), (1,1), (1,5), (2,7), (2,3), (2,1), (2,5)\}$

$R(E \times F) = \{(3,0), (3,2), (3,4), (3,1), (7,0), (7,2), (7,4), (7,1), (9,0), (9,2), (9,4), (9,1)\}$

And so on.

Representation of Functions

- 1. **Explicitly** by assigning each element of A to an element in B directly.

Example:

$A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$
 $f: A \rightarrow B$:

- o $f(1) = 4$
- o $f(2) = 5$
- o $f(3) = 6$

- 2. **Formula** by assigning the image as a formula that depends on the value given to a variable used.

Example:

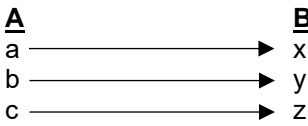
$A = \{1, 2, 3, 4\}$, $B = \{1, 4, 9, 16\}$
 $f(x) = x^2$, where each element in A is mapped to its square in B. Therefore,

- o $f(1) = 1$
- o $f(2) = 4$
- o $f(3) = 9$
- o $f(4) = 16$

- 3. **Graphically** using arrow diagrams, the following are different representations of functions using arrow diagrams.

Example:

$A = \{a, b, c\}$, $B = \{x, y, z\}$



Types of Functions

One-to-One Function or Injective

Functions that never assign the same values to two or more different elements in the **domain** are said to be **one-to-one**.

Meaning, a function is one-to-one (denoted 1-1) or injective if preimages are unique. Each value in the range corresponds to **exactly one value** in the domain.

Example:

$A = \{1, 2, 3, 4\}$, $B = \{5, 6, 7, 8, 9, 10\}$
 $f: A \rightarrow B$ defined as $f(x) = x + 4$. Therefore,

- o $f(1) = 5$
- o $f(2) = 6$
- o $f(3) = 7$
- o $f(4) = 8$

The above function is considered injective, as each element in A has a **unique** image in B, and no two elements in A share the same output.

Onto Function or Surjective

Functions that maps every element in the **co-domain** to at least one element in the **domain** is an onto function.

Meaning, is onto or surjective if every y in B has a pre-image. Every element in the codomain maps to **at least one element** in the domain.

Example:



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$A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$
 $f: A \rightarrow B$ defined as:

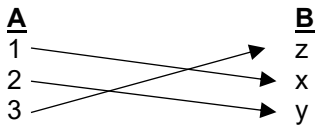
- $f(1) = a$
- $f(2) = b$
- $f(3) = c$
- $f(4) = a$

In the above example, every element in B has at least one pre-image from A. The element "a" has two pre-images (1 and 4), but all elements in B are mapped, making the function onto.

Bijjective Function

Functions that are both injective and surjective are considered bijective.

Example:
 $A = \{1, 2, 3\}$, $B = \{z, x, y\}$



From the given function, it's considered a **one-to-one** function since every element in the domain {1,2,3} is assigned a unique element in the co-domain.

It's also an **onto** function since the co-domain {w,x,y} is equal to the range which means all values were mapped to the domain. Therefore, the function is considered bijective.

We can also conclude that whenever there is a bijection from A to B, the two sets must have the same number of elements or the same cardinality.

Function Inverse and Composition
Inverse of a Function

The inverse of a function **f**, denoted as **f⁻¹**, reverses the mapping of **f**, meaning it swaps the roles of the domain and range. It can be defined as

$f^{-1}(y) = x \text{ iff } f(x) = y$

A function must be **bijective** (both one-to-one and onto) to have an inverse.

Examples:
1. What will be the inverse of $f: R \rightarrow R, f(x) = 2x - 3$?

When the function is represented with formula, to get the inverse, we solve for x using transposition. We replace domain variable x (in the function) with y to differentiate the formula, as seen here:

$$\begin{aligned} f^{-1}(y) &= 2x - 3 \\ y &= 2x - 3 \\ y + 3 &= 2x \\ \frac{y + 3}{2} &= x \end{aligned}$$

Therefore, the inverse function is:

$$f^{-1}(x) = \frac{x + 3}{2}$$

If x is assigned a specific value, for instance 4, we replace the value of x in the formula.

$$\begin{aligned} f(4) &= 2x - 3 \\ &= 2(4) - 3 \\ &= 8 - 3 \\ &= 5 \end{aligned}$$

Therefore, $f^{-1}(5) = 4$. To prove this, let us try replacing y with 5 in the formula we derived earlier.

$$\begin{aligned} x &= \frac{y + 3}{2} \\ x &= \frac{5 + 3}{2} \\ x &= \frac{8}{2} \\ x &= 4 \end{aligned}$$

From the definition $f^{-1}(y) = x \text{ iff } f(x) = y$, is proven here where $f^{-1}(5) = 4 \text{ iff } f(4) = 5$. This should be the case for all values of x and y.

2. What will be the inverse of $f: R \rightarrow R, f(x) = 3x - 2$?

$$\begin{aligned} f^{-1}(y) &= 3x - 2 \\ y &= 3x - 2 \\ y + 2 &= 3x \\ \frac{y + 2}{3} &= x \end{aligned}$$

Therefore, the inverse function is:

$$f^{-1}(x) = \frac{x + 2}{3}$$

To check, replace the value of the inverse function (which we have solved above), like such:

$$\begin{aligned} f^{-1}(7) &= \frac{7 + 2}{3} \\ &= \frac{9}{3} \\ &= 3 \end{aligned}$$

Then use the output of the inverse function as an input to the original function, like such:

$$\begin{aligned} f(3) &= 3(3) - 2 \\ &= 9 - 2 \\ &= 7 \end{aligned}$$

Composition of Functions

Function composition is a process where two functions are combined to form a new function. In simpler terms, you take the output of one function and use it as the input for another function.

Simply put, the composition of two functions *f* and *g*, denoted as $(f \circ g)(x)$, means applying *g(x)* first, then applying *f* to the result.



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If f and g are one-to-one, then the function $(f \circ g / g \circ f)$ is also one-to-one.
If f and g are onto, then the function $(f \circ g / g \circ f)$ is also onto.

Examples:
1. Let $f(x) = x^2$ and $g(x) = x + 1$.
a) What is $(f \circ g)(x)$?

$f(g(x)) = f(x + 1)$ Replace the value of x in $f(x)$ with the equation $g(x)$,
 $= (x + 1)^2$
 $= (x + 1)(x + 1)$ Then simplify the resulting equation. In this case, the FOIL method is applicable.

$f(g(x)) = x^2 + 2x + 1$

b) What is $(g \circ x)(3)$?

$g(f(x)) = g(x^2)$ Replace the value of x in $g(x)$ with the equation $f(x)$
 $= (x^2) + 1$ Once the equation reaches its simplest form, then replace the value of x with the given value, which is 3.
 $= (3^2) + 1$
 $= 10$

Relations

Relationships between elements of sets are represented using the structure called a **relation (R)**, which is just a subset of the Cartesian product of the sets.

The most direct way to express a relationship between elements of two sets is to use **ordered pairs** made up of two related elements. For this reason, sets of ordered pairs are called **binary relations**.

Let **A** and **B** be two sets. A **binary relation R** from A to B, written $R: A \times B$ is a subset of $A \times B$ or $R \subseteq A \times B$.

- $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$
- The notation $a R b$, a is related to b by R , means that $(a, b) \in R$.
- $a \not R b$ is used to denote $(a, b) \notin R$.

Relations can be used to solve problems such as determining which pairs of cities are linked by airline flights in a network, finding a viable order for the different phases of a complicated project, or producing a useful way to store information in computer databases.

Let A, B be any sets. A **binary relation R** from A to B , written as $R: A \times B$ is a subset of the set $A \times B$.

For example.
Let $A = \{0, a, b\}$ and $B = \{a, b\}$.

Then $\{(0, a), (0, b), (a, a), (b, b)\}$ is a relation from A to B . This means, for instance, that $0 R a$. But that $a \not R b$, a is not related to b .

Functions as Relations

- **Relation:** A relation from A to B is simply a set of ordered pairs that could be any subset of $A \times B$, and there is no restriction on how many pairs from A can relate in B . For example, $(1, a)$ and $(1, b)$ can both be in a relation, which means 1 is related to both a and b .
- **Function:** A function is a special kind of relation. It's a relation where every element of A must be related to exactly one element of B . In other words, no element of A can have more than one corresponding element in B .

In simple terms, all functions are relations, but not all relations are functions.

Examples:
1. Let $A = \{1, 2, 3\}, B = \{0, 1, 2, 4\}$. Find $R = \{(a, b) \mid a = b\}$.

First, solve for the Cartesian product of the two sets:

$A \times B = \{(1, 0), (1, 1), (1, 2), (1, 4), (2, 0), (2, 1), (2, 2), (2, 4), (3, 0), (3, 1), (3, 2), (3, 4)\}$

Among the given ordered pairs, $(1, 1)$ and $(2, 2)$ satisfies the condition. Therefore,

$R = \{(1, 1), (2, 2)\}$.

Thus, $R \subseteq A \times B$, as all elements of R can be found in $A \times B$.

2. Let $A = \{1, 2, 3, 4, 5, 6\}$. Find $R = \{a, b \mid a \text{ divides } b\}$.

The Cartesian product $A \times A$ consists of all ordered pairs where both elements are from A . Therefore,

$A \times A = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$

Among these pairs, the relation R includes only the pairs where the first element divides the second element without a remainder. Therefore,

$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$

Properties of Relations
Reflexive

A relation R on set A is called reflexive if $(a, a) \in R$ for every element $a \in A$. This means that if every element of set A is an

Example 1:
Let $A = \{1, 2, 3, 4, 5, 6\}$ and define the relation R as:
 $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (1, 2), (2, 4)\}$



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Since $(1, 1), (2, 2), (3, 3), (4, 4), (5, 5)$, and $(6, 6)$ are elements of R , every element in A is related to itself. Thus, R is reflexive.

Example 2:

Consider the following relations on $\{a,b,c,d\}$. Which among these relations are reflexive?

- a. $R_1 = \{(a, a), (a, b), (b, c), (c, d), (d, a)\}$
- b. $R_2 = \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a)\}$
- c. $R_3 = \{(a, a), (b, b), (c, c), (d, d)\}$
- d. $R_4 = \{(a, b), (b, c), (c, d)\}$
- e. $R_5 = \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a), (b, c), (c, b)\}$

Answers:

- a. **Not reflexive** because $(b,b), (c,c)$, and (d,d) are missing.
- b. **Reflexive** because $(a,a), (b,b), (c,c)$, and (d,d) are included.
- c. **Reflexive** since all elements relate to themselves.
- d. **Not reflexive** because none of $(a,a), (b,b), (c,c)$, or (d,d) are present.
- e. **Reflexive** since $(a,a), (b,b), (c,c)$, and (d,d) are included.

Symmetric

A relation R on a set A is called **symmetric** if $(b, a) \in R$ holds when $(a, b) \in R$ for all $a, b \in A$.

In other words, a relation is **symmetric** whenever (a, b) is in R , and (b, a) must also be in R .

Example:

Consider the following relations on $\{a, b, c, d\}$. Which among these relations are symmetric?

- $R_1 = \{(a, a), (a, b)\}$
- $R_2 = \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a)\}$
- $R_3 = \{(a, a), (b, b), (c, c), (d, d)\}$
- $R_4 = \{(a, b), (b, c), (c, d)\}$
- $R_5 = \{(a, b), (b, a), (b, c), (c, b)\}$

Answers:

- a. **Not symmetric.** The pair (a,b) is in R_1 , but its symmetric pair (b,a) is missing.
- b. **Symmetric.** Every pair (a,b) in R_2 has its corresponding symmetric pair (b,a) .
- c. **Symmetric.** R_3 contains only self-pairs (a,a) , which always satisfy symmetry.
- d. **Not symmetric.** All pairs in R_4 are missing their symmetric pair.
- e. **Symmetric.** Every pair (a,b) in R_5 has its corresponding symmetric pair (b,a) .

Antisymmetric

The relation R on the set A is antisymmetric if $\forall a \forall b ((a, b) \in R \wedge (b, a) \in R) \rightarrow (a = b)$.

In simpler terms, a relation R on a set A is **antisymmetric** if:

If (a, b) is in R and (b, a) is also in R , then a must be equal to b .

Examples:

Consider the following relations on $\{a, b, c, d\}$. Which among these relations are antisymmetric?

- a. $R_1 = \{(a, a), (a, b), (b, c), (c, d)\}$
- b. $R_2 = \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a)\}$
- c. $R_3 = \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, c), (c, d)\}$
- d. $R_4 = \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a), (b, c), (c, b)\}$
- e. $R_5 = \{(a, a), (b, b), (c, c), (d, d)\}$

Answers:

- a. **Antisymmetric.** There are no pairs (a, b) and (b, a) where $a \neq b$.
- b. **Not antisymmetric.** The pair (a, b) is in R_2 , and its reverse (b, a) is also in R_2 , but $a \neq b$.
- c. **Antisymmetric.** No two different elements are related in both directions.
- d. **Not antisymmetric.** The pair (b, c) is in R_4 , and (c, b) is also in R_4 , but $b \neq c$.
- e. **Antisymmetric.** The relation only contains pairs that are identical, which are reflexive and do not violate antisymmetry.

Transitive

A relation R on set A is called **transitive** if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$ for all $a, b, c \in A$.

In simple terms, a relation is **transitive** if whenever a is related to b , and b is related to c , then a must also be related to c .

Examples:

Consider the following relations on $\{a, b, c, d\}$. Which among these relations are transitive?

- a. $R_1 = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,3), (1,3)\}$
- b. $R_2 = \{(1,2), (2,3), (3,4)\}$
- c. $R_3 = \{(1,1), (2,2), (3,3), (4,4), (1,3), (3,4)\}$
- d. $R_4 = \{(1,2), (2,1), (3,4), (4,3)\}$
- e. $R_5 = \{(1,2), (2,3), (1,3), (2,4), (1,4)\}$

Answers:

- a. **Transitive.** Since $(1,2)$ and $(2,3)$ exist, $(1,3)$ must be in R_1 —and it is. No missing pairs that break transitivity.
- b. **Not transitive.** $(1,2)$ and $(2,3)$ exist, but $(1,3)$ is missing. Similarly, $(2,3)$ and $(3,4)$ exist, but $(2,4)$ is missing.
- c. **Not transitive.** $(1,3)$ and $(3,4)$ exist, but $(1,4)$ is missing.
- d. **Not transitive.** $(1,2)$ and $(2,1)$ exist, but $(1,1)$ is missing. Similarly, $(3,4)$ and $(4,3)$ exist, but $(3,3)$ is missing.
- e. **Transitive.** Since $(1,2)$ and $(2,3)$ are present, $(1,3)$ must also be in R_5 —and it is. Similarly, $(1,4)$ and $(2,4)$ imply $(1,4)$ is present, and it is. Every necessary pair is included, making it transitive.

Combining Relations

Because relations from A to B are subsets of $A \times B$, two relations from A to B can be combined in any way two sets can be combined.

Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$. The relations $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ and $R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$ can be combined to obtain:

- a) $R_1 \cup R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\}$
- b) $R_1 \cap R_2 = \{(1, 1)\}$



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- c) $R_1 - R_2 = \{(2,2),(3,3)\}$
- d) $R_2 - R_1 = \{(1,2), (1,3), (1,4)\}$

Composite of Two Relations

Let R be a relation from a set A to a set B and S a relation from B to a set C . The composite of R and S is the relation consisting of ordered pairs (a,c) , where $a \in A, c \in C$, and for which there exists an element $b \in B$ such that $(a,b) \in R$ and $(b,c) \in S$.

We denote the composite of R and S by $S \circ R$. Computing the composite of two relations requires that we find elements that are the second element of ordered pairs in the first relation and the first element of ordered pairs in the second relation.

Example 1:
What is the composite of the relations R and S , where:
 R is the relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with
 $R = \{(1,1), (1,4), (2,3), (3,1), (3,4)\}$,

S is the relation from $\{1, 2, 3, 4\}$ to $\{0, 1, 2\}$ with
 $S = \{(1,0), (2,0), (3,1), (3,2), (4,1)\}$?

To compute the composite of the relations $S \circ R$, we need to understand the basic procedure for composing two relations.

The composite of relations $S \circ R$ is defined as follows: for any pair (a, c) in $S \circ R$, there exists an intermediate element b such that:

- (a, b) is in R
- (b, c) is in S

In other words, we need to find all pairs (a, c) such that the second element of a pair in R matches the first element of a pair in S .

We match the second element of each pair in R with the first element of a pair in S and form new pairs (a, c) .

R Pair	Matching S Pair	Resulting Pair in $S \circ R$
(1,1)	(1,0)	(1,0)
(1,4)	(4,1)	(1,1)
(2,3)	(3,1)	(2,1)
(2,3)	(3,2)	(2,2)
(3,1)	(1,0)	(3,0)
(3,4)	(4,1)	(3,1)

Therefore, the composite relation $S \circ R$ is:
 $S \circ R = \{(1,0), (1,1), (2,1), (2,2), (3,0), (3,1)\}$

Composing a Relation with Itself

Let R be a relation on the set A . The powers of R are defined as:

- $R^1 = R$
- $R^2 = R \circ R$
- $R^3 = R^2 \circ R$ and so on.

For a pair (a, c) to be in R^2 , there must be some element b that connects them:

- (a, b) is in R
- (b, c) is also in R
- Then, we add (a, c) to R^2

Example:

Given R on $A = \{1,2,3,4\}$: $R = \{(1,1), (2,1), (3,2), (4,3)\}$, to find $R^2 = R \circ R$.

Step 1: Start with each pair in R
Each pair in R is written as (a, b) :

- $(1,1)$
- $(2,1)$
- $(3,2)$
- $(4,3)$

Step 2: Find matches in R
For each pair (a, b) in R , find another pair (b, c) in R where the b 's match.

First Pair (a, b) in R	Matching Pair (b, c) in R	Result (a, c) added to R^2
(1,1)	(1,1)	(1,1)
(2,1)	(1,1)	(2,1)
(3,2)	(2,1)	(3,1)
(4,3)	(3,2)	(4,2)

Matrices

Matrices are used throughout discrete mathematics to express relationships between elements in sets. They are also used for representing relations, specifically zero-one matrices. They are generally appropriate for representation of relations in computer programs.

A **matrix** is a rectangular array of objects (usually numbers).
An $m \times n$ ("m by n") matrix has exactly m horizontal rows, and n vertical columns.
An $n \times n$ matrix is called a square matrix, whose order or rank is n .

Matrix Equality

Two matrices **A** and **B** are considered equal iff they have the same number of rows, the same number of columns, and all their corresponding elements are equal.

These matrices are not equal, they have the same number of rows but different number of columns, with differing values.

$$\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix} \neq \begin{bmatrix} 3 & 2 & 0 \\ -1 & 6 & 0 \end{bmatrix}$$

The rows in a matrix are usually indexed 1 to m from **top to bottom**. The columns are usually indexed 1 to n from **left to right**. Elements are indexed by row, then column.

$$A = [a_{i,j}] = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$$

Matrix Sum

The **sum** $A+B$ of two matrices **A**, **B** (which **must** have the same number of rows, and the same number of



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columns) is the matrix (also with the same shape) given by adding corresponding elements of **A** and **B**.

$$\begin{bmatrix} 2 & 6 \\ 0 & -8 \end{bmatrix} + \begin{bmatrix} 9 & 3 \\ -11 & 3 \end{bmatrix} = \begin{bmatrix} 11 & 9 \\ -11 & -5 \end{bmatrix}$$

Both matrices have the same size of 2x2, the resulting matrix with the same size, contains the sums of the specific values on the same position in each matrix.

Matrix Multiplication

Matrices **M** and **N** can be multiplied if **M** is a **m x k** matrix and **N** is a **k x n** matrix. The resulting product matrix will be an **m x n** matrix.

Multiplication of matrices only applies when the first matrix's number of columns is equal to the second matrix's number of rows. With the order of the matrices explicitly stated, we can say that Matrix multiplication is **not** commutative. **AB** is not equal to **BA**.

Assume that **A** is **2x3** matrix and **B** is a **3x4** matrix, the product of **AB** will be a **2x4** matrix. While **BA** is not even possible, since **B** has 4 columns and **A** has 2 rows which does not satisfy the condition for matrix products.

Example 1:

Given the following matrices, with sizes of 2x3 and 3x2 respectively, the matrix product will be 2x2 in size. To get the value on the 1st row and 1st column, we get the sum of products of the 1st row in **A** with the 1st column in **B**, as seen on the figure.

To work out the answer for the 1st row and 1st column:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & \dots \\ \dots & \dots \end{bmatrix}$$

The "Dot Product" is where we multiply matching members, then sum up:

$$(1, 2, 3) \bullet (7, 9, 11) = 1 \times 7 + 2 \times 9 + 3 \times 11 = 58$$

For the 1st row and 2nd column, we repeat the process using 1st row of **A** and 2nd column of **B**.

Want to see another example? Here it is for the 1st row and 2nd column:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & 64 \\ \dots & \dots \end{bmatrix}$$

$$(1, 2, 3) \bullet (8, 10, 12) = 1 \times 8 + 2 \times 10 + 3 \times 12 = 64$$

For the last two values,

We can do the same thing for the 2nd row and 1st column:

$$(4, 5, 6) \bullet (7, 9, 11) = 4 \times 7 + 5 \times 9 + 6 \times 11 = 139$$

And for the 2nd row and 2nd column:

$$(4, 5, 6) \bullet (8, 10, 12) = 4 \times 8 + 5 \times 10 + 6 \times 12 = 154$$

And we get:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}$$

Example 2:

A	B	AB
$\begin{bmatrix} 0 & 1 & -1 \\ 2 & 0 & 3 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 & 1 & 0 \\ 2 & 0 & -2 & 0 \\ 1 & 0 & 3 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & -5 & -1 \\ 3 & -2 & 11 & 3 \end{bmatrix}$

In the example above, **A** is a 2x3 matrix and **B** is a 3x4 matrix, resulting to a 2x4 matrix for **AB**.

Identity Matrices

An **identity matrix** is a square matrix (same number of rows and columns) where:

- The **diagonal** (from top-left to bottom-right) has **1's**
- All other positions have **0's**

For example, a 3x3 identity matrix looks like this:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The identity matrix is called this because when you multiply it by any matrix, it doesn't change that matrix. It's like multiplying a number by 1 in regular math—it keeps the number the same.

Matrix Inverses

For some (but not all) square matrices **A**, there exists a unique multiplicative **inverse** **A⁻¹** of **A**, a matrix such that **A⁻¹A = I_n**. If the inverse exists, it is unique, and **A⁻¹A = AA⁻¹**. Refer to the example:

for a 2x2 matrix the inverse is:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

↑
determinant

Let us try an example:

$$\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}^{-1} = \frac{1}{4 \times 6 - 7 \times 2} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$$



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So, let us check to see what happens when we multiply the matrix by its inverse:

$$\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix} = \begin{bmatrix} 4 \times 0.6 + 7 \times -0.2 & 4 \times -0.7 + 7 \times 0.4 \\ 2 \times 0.6 + 6 \times -0.2 & 2 \times -0.7 + 6 \times 0.4 \end{bmatrix} \\ = \begin{bmatrix} 2.4 - 1.4 & -2.8 + 2.8 \\ 1.2 - 1.2 & -1.4 + 2.4 \end{bmatrix} \\ = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The given is a square matrix, the inverse denoted by the -1 power, is computed by first rearranging the values and changing the signs accordingly. Then multiply each value in the transformed matrix with $1/\text{determinant}$ computed as ad-bc .

By multiplying the matrix with its inverse, the product is an identity matrix, where the values in the diagonal are 1s. Therefore, this square matrix has a multiplicative inverse as defined $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$.

So, let us check to see what happens when we multiply the matrix by its inverse:

$$\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix} = \begin{bmatrix} 4 \times 0.6 + 7 \times -0.2 & 4 \times -0.7 + 7 \times 0.4 \\ 2 \times 0.6 + 6 \times -0.2 & 2 \times -0.7 + 6 \times 0.4 \end{bmatrix} \\ = \begin{bmatrix} 2.4 - 1.4 & -2.8 + 2.8 \\ 1.2 - 1.2 & -1.4 + 2.4 \end{bmatrix} \\ = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Powers of Matrices

If $\mathbf{A}$ is an $n \times n$ square matrix and $p \geq 0$, then:
 $\mathbf{A}^p$: $\mathbf{A}\mathbf{A}\mathbf{A}\cdots\mathbf{A}$ where p (power) refers to the number of times the matrix is multiplied to itself.

Example:

$$\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}^3 = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \\ = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 3 & 2 \\ -2 & -1 \end{bmatrix} \\ = \begin{bmatrix} 4 & 3 \\ -3 & -2 \end{bmatrix}$$

The matrix is multiplied to itself three times, to compute we apply the steps for product matrix for two matrices at a time. The resulting product matrix is then multiplied again to the given matrix.

Process:
Step 1: Compute $\mathbf{A}^2$ (the square of the matrix)

$$\mathbf{A}^2 = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \times \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$$

Multiply the matrices by applying the matrix multiplication rule:

$$\mathbf{A}^2 = \begin{pmatrix} (2 \times 2 + 1 \times -1) & (2 \times 1 + 1 \times 0) \\ (-1 \times 2 + 0 \times -1) & (-1 \times 1 + 0 \times 0) \end{pmatrix} \\ \mathbf{A}^2 = \begin{pmatrix} (4 - 1) & (2 + 0) \\ (-2 + 0) & (-1 + 0) \end{pmatrix} \\ \mathbf{A}^2 = \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix}$$

Step 2: Compute $\mathbf{A}^3$ (multiply $\mathbf{A}^2$ by $\mathbf{A}$)
Now, calculate $\mathbf{A}^3 = \mathbf{A}^2 \times \mathbf{A}$:

$$\mathbf{A}^3 = \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix} \times \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$$

Perform the matrix multiplication:

$$\mathbf{A}^3 = \begin{pmatrix} (3 \times 2 + 2 \times -1) & (3 \times 1 + 2 \times 0) \\ (-2 \times 2 + -1 \times -1) & (-2 \times 1 + -1 \times 0) \end{pmatrix} \\ \mathbf{A}^3 = \begin{pmatrix} (6 - 2) & (3 + 0) \\ (-4 + 1) & (-2 + 0) \end{pmatrix} \\ \mathbf{A}^3 = \begin{pmatrix} 4 & 3 \\ -3 & -2 \end{pmatrix}$$

Matrix Transposition

The **transpose** of a matrix $\mathbf{A}$, written as $\mathbf{A}^T$, is formed by swapping its rows and columns.

If $\mathbf{A}$ is an $m \times n$ matrix: $\mathbf{A} = [a_{ij}]$

Then its transpose $\mathbf{A}^T$ is an $n \times m$ matrix: $\mathbf{A} = [a_{ji}]$

This means that the element in row i , column j of $\mathbf{A}$ moves to row j , column i in $\mathbf{A}^T$.

Examples:

$$\begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & -2 \end{bmatrix}^T = \begin{bmatrix} 2 & 0 \\ 1 & -1 \\ 3 & -2 \end{bmatrix} \quad \begin{bmatrix} 6 & 4 & 24 \\ 1 & -9 & 8 \end{bmatrix}^T = \begin{bmatrix} 6 & 1 \\ 4 & -9 \\ 24 & 8 \end{bmatrix}$$

If the transpose of a matrix is equal to the given matrix, the matrix is considered **symmetric**.

It is impossible for a matrix with different number of rows and columns to be symmetric, as seen on the previous examples with size of 2×3 , the transpose was 3×2 which makes them unequal. Therefore, only square matrices may be symmetric, but not all.

$$\begin{bmatrix} -2 & 1 & 3 \\ 1 & 0 & -1 \\ 3 & -1 & 2 \end{bmatrix}^t = \begin{bmatrix} -2 & 1 & 3 \\ 1 & 0 & -1 \\ 3 & -1 & 2 \end{bmatrix}$$

Zero-One Matrices

As mentioned in the introduction to matrices, this type of matrices is generally used for the representation of relations in computer programming.



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They have the same properties as any matrix but only have 0s and 1s as values. Meaning:

- 0 = False
- 1 = True

Example:

Find the **join** and **meet** of the zero-one matrices A and B.

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

A. The **join** of A, B (both $m \times n$ zero-one matrices)

Each element in the same position for each matrix are combined using **disjunction**, 0 when both are 0s and 1 otherwise. Therefore,

$$A \vee B = \begin{bmatrix} 1 \vee 0 & 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 1 \vee 1 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

B. The **meet** of A, B

Each element in the same position for each matrix are combined using **conjunction**, 1 when both are 1s and 0 otherwise.

The meet of A and B is

$$A \wedge B = \begin{bmatrix} 1 \wedge 0 & 0 \wedge 1 & 1 \wedge 0 \\ 0 \wedge 1 & 1 \wedge 1 & 0 \wedge 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Boolean Products

Let $A=[a_{ij}]$ be an $m \times k$ zero-one matrix, and let $B=[b_{ij}]$ be a $k \times n$ zero-one matrix,

The *boolean product* of A and B is like normal matrix product, but instead of adding the products, **disjunction** is used instead to combine the corresponding conjunction of the Boolean values.

Examples:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

$$\begin{aligned} A \odot B &= \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix} \\ &= \begin{bmatrix} 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \\ 0 \vee 0 & 0 \vee 1 & 0 \vee 1 \\ 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}. \end{aligned}$$

Each row in A is combined with every row in B, where every element is combined with conjunction and the results are combined using disjunction. In relation to matrix product, the **conjunction** operator replaces

multiplication of the values and **disjunction** replaces **addition** of the products.